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Encapsulated fixed window module for a motor vehicle

Abstract

An encapsulated fixed window module for a motor vehicle includes a fixed window pane, a first division post profile defining a longitudinal groove configured to receive a guide carried by a movable window pane, and having an exterior wall facing a fixed pane interior surface. An over-molded material secures the first division post profile to the fixed window pane, the over-molded material includes a first over-molded portion extending between the division post exterior wall and the fixed pane interior surface and a second over-molded portion extending between the lateral wall and the interior surface. The first over-molded portion has a first thickness which is constant along a major portion of the first over-molded portion. The second over-molded portion has a second thickness which is constant along a major portion of the second portion due to a co-extruded feature on the first division post profile.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates generally to an encapsulated fixed window module for a motor vehicle and such a motor vehicle.

BACKGROUND OF THE INVENTION

(2) In a number of motor vehicles, a door assembly is designed such that there is a first, e.g., forward, window opening that carries a movable window pane and a second, e.g., rearward, opening that is equipped with a fixed or stationary window pane. Of course, the forward window

pane may be fixed and the rearward window pane may be moveable. Fixed windows of conventional design may be assembled from discrete elements that include the window panel, a post that serves as a track for an adjacent movable window and various configurations of moldings or trim pieces or division posts, most specifically an extruded header/B-pillar trim piece that may be mechanically attached to the window module assembly using an interlocking channel or the like.

(3) The division post in some prior art designs comprises an article in the nature of a pre-formed resilient band. In those designs, the window is inserted into a channel of the trim band which resiliently engages the window pane, but often less than securely. Conventional designs suffer from a number of deficiencies due to their assembly from discrete components such as water leakage, noise from the exterior (wind noise) or the interior, and problems generally with fit and finish. In addition, the assembly of these modules is often difficult, particularly where the individual components are manufactured by more than one supplier.

(4) A few of the problems of conventional designs are overcome through the use of over-molding processes by which a portion of the trim or post surrounding the window pane is fabricated by encapsulating the window periphery with a polymer using injection molding techniques. In essence, the window pane is placed in a mold cavity which is configured to produce a circumferential trim at the pane edges. A division post profile and a header profile are also placed in the mold cavity. In some case, an additional margin seal extrusion can be added to the mold cavity. The edges of the pane on which the injection molded trim will adhere are cleaned and primed to promote adhesion and to create a water-impervious seal in the formed article. The mold is then closed and the material is injected such that the window pane edges are encapsulated to form the desired trim geometry.

(5) FIG. 1 is a cross sectional view of adjacent window panes **1**, **2** of a motor vehicle. The window pane **1** is fixed and the window pane **2** is movable. The fixed window pane **1** has an exterior surface **1a**, an interior surface **1b**, and a peripheral edge **1c** extending between said exterior and interior surfaces **1a**, **1b**. A division post **3** extends along the peripheral edge **1c** and is secured thereto by an over-molded material **4**.

(6) The division post **3** is U-shaped in cross section and has an exterior wall **3a**, an interior wall **3b**, and a lateral wall **3c** connecting said exterior and interior walls **3a**, **3b**. The division post **3** defines a longitudinal groove **5** configured to receive a portion of the movable window pane **2**. The lateral wall **3c** of the division post **3** is in abutment against the peripheral edge **1c** of the fixed window pane **1** and said over-molded material **4** comprises a first portion **4a** extending between the lateral wall **3c** and the exterior surface **1a**, and a second portion **4b** extending between the lateral wall **3c** and the interior surface **1b**.

(7) This configuration creates a small offset **O** between the interior surface **1b** and the interior wall **3b** and requires limited amount of over-molded material on both sides of the fixed window pane **1**. The risk of appearance issue due to sink marks is therefore low.

(8) The exterior appearance of the window modules is very important for the customer, i.e., the user of the motor vehicle. A solution for enhancing the appearance of a window module is to provide the periphery of the window pane with finished panels which are located outside the motor vehicle and are visible by the user. Said finished panels are for instance esthetic metal components which are secured to the window module.

(9) It is also known to enhance the esthetic appearance of a window module by lying flush the outside of its stationary window pane with the outside of an adjacent movable window pane.

(10) FIG. 2 is a cross sectional view of adjacent and flush window panes **1**, **2** of a motor vehicle. The division post **3** extends along the peripheral edge **1c** and is secured to the interior surface **1b** by an over-molded material **4**.

(11) The division post **3** defines a longitudinal groove **5** configured to receive a guiding mean **6** carried by the movable window pane **2**. The exterior wall **3a** of the division post **3** faces the interior surface **1b** and is secured thereto by the over-molded material **4**.

(12) The over-molded material **4** comprises a first portion **4a** extending between the exterior wall **3a** and the interior surface **1b** and a second portion **4b** extending between the lateral wall **3c** and the interior surface **1b**. FIG. 2 shows that the first portion **4a** has a thickness E1 which is constant (the thickness, E1 being measured in a direction D1 perpendicular to the interior surface **1b** and/or to the exterior wall **3a**). On the contrary, the second portion **4b** has a bigger thickness E2 which is not constant and which varies (the thickness, E2 is measured in a direction D2 perpendicular to the lateral wall **3c**). The thickness E2 of the second portion **4b** is higher in the proximity of the interior surface **1b** than on the opposite end in the proximity of the interior wall **3b**.

(13) The over-molded material **4** comprises an outer face **4c** which extends between the lateral wall **3c** and the interior surface **1b** and which includes a major portion **4c1** which is inclined with respect to the interior surface **1b**. Due to feasibility reasons, it is not possible to get a major portion **4c1** perpendicular to the interior surface **1b** during the over-molding process.

(14) In this flush configuration, a large offset O is created between the interior surface **1b** and the interior wall **3b** and a large amount of over-molded material **4** is needed to fill this offset O. The risk of appearance issue due to sink marks is therefore high. Indeed, after de-molding, the over-molded material **4** cools and hardens. The large amount of over-molded material promotes the formation of sink marks and surface defects on the outer face **4c**. Since this outer face **4c** may be visible to the user of the motor vehicle, these sink marks and surface defects are problematic.

(15) The present invention provides at least an encapsulated fixed window module which can be used in a flush configuration and which avoids the above-mentioned appearance issue of the over-molded material.

SUMMARY OF THE INVENTION

(16) According to the invention there is provided an encapsulated fixed window module for a motor vehicle, comprising: a fixed window pane having an exterior surface, an interior surface, and a peripheral edge extending between said exterior and interior surfaces, at least one division post extending along at least a portion of said peripheral edge, said division post comprising a first longitudinal profile which is U-shaped in cross section and which has an exterior wall, an interior wall, and a lateral wall connecting said exterior and interior walls, said first profile defining a longitudinal groove configured to receive a guiding mean carried by a movable window pane, said exterior wall facing said interior surface, and an over-molded material securing said first profile to said fixed window pane, said over-molded material comprising a first portion extending between said exterior wall and said interior surface and a second portion extending between said lateral wall and said interior surface, said first and second portions being connected one to the other, said first portion having a first thickness which is constant along a major portion of said first portion, said second portion having an outer face which extends between the lateral wall and said interior surface and which includes a major portion which is inclined with respect to said interior surface, wherein said second portion has a second thickness which is constant along a major portion of said second portion.

(17) In the present specification, the words “inner”, “inside”, “interior”, etc., make reference to the inside of a motor vehicle. The words “outer”, “outside”, “exterior”, etc., make reference to the outside of the motor vehicle. Then, an outer element is located at the outer side of the vehicle. A first outer element, portion or surface may be visible by a user of the motor vehicle or may be hidden by a second outer element, portion or surface covering said first outer element.

(18) In the present specification, “aligned” and “substantially aligned” mean that two elements are strictly aligned or are aligned in a range 0 to 5 mm, preferably 0 to 3 mm and more preferably 0 to 1 mm.

(19) The present invention provides a solution for enhancing the esthetic appearance of a fixed window module. This aim is achieved by keeping the thickness of the over-molded material as constant as possible. The second portion of the over-molded material is constant along a major portion of this second portion along a direction perpendicular to the interior surface of the fixed

window pane. The thicknesses of the first and second portions of the over-molded material may be similar. Having a consistent thickness allows all the molded areas to cool at the same speed and prevent “sink-marks” and other appearance defects.

(20) The encapsulated fixed window module according to the invention may comprise one or more of the following features, taken alone from each other or in combination with each other: said lateral wall comprises a tooth which includes a first surface parallel to said interior surface, and a second surface which is inclined with respect to said interior surface, said first portion extending between said first surface and said interior surface and said second portion extending between said second surface and said interior surface; said first profile is extruded; said tooth is co-extruded with said first profile; said first surface is planar and said second surface is planar or curved; said second surface is concave or convex; said second surface is mainly parallel to said outer surface or provides a constant offset when this second surface is concave or convex; said first and second surfaces are V-shaped in cross section; said tooth has a tip having an angle comprises between 20 and 85°, and preferably between 50 and 70°; said over-molded material is V-shaped; said over-molded material includes a protruding rib extending on the interior surface; said exterior wall comprises a longitudinal end opposite said lateral wall, said longitudinal end comprising a protruding leg which is perpendicular to the exterior wall and is in abutment against said peripheral edge; said protruding leg carries a first sealing lip which is configured to abut against a peripheral edge of said movable window pane; said longitudinal end and/or said exterior wall carries a protruding lip which abuts against said interior face and which is in contact with said over-molded material; said protruding lip is advantageous because it is used as a mold shut off between the fixed pane and the first profile; an empty space is defined between said fixed window pane on one hand, and said protruding leg and lip on the other hand; $0.8 \cdot E_1 \leq E_2 \leq 1.2 \cdot E_1$; and preferably $0.9 \cdot E_1 \leq E_2 \leq 1.1 \cdot E_1$ and more preferably $E_1 = E_2$; E_1 and E_2 being respectively said first and second thicknesses; said first profile comprises an embedded metallic rail which includes a leg extending outwardly and facing said peripheral edge; the thicknesses of the first and second portions are constant along a major portion of a length of the first profile; the thickness of the first portion is measured in a direction which is perpendicular to the interior surface of the fixed window pane; this direction may be perpendicular to said exterior wall or to a surface of said exterior wall; the thickness of the second portion is measured in a direction which is perpendicular to said lateral wall or to a surface of said lateral wall; the thickness of the first portion is substantially constant when the first profile is viewed in cross section; the thickness of the second portion is constant when the first profile is viewed in cross section; said over-molded material does not cover said peripheral edge along said first profile; said exterior wall of said first profile is applied onto said interior surface; said exterior wall is L-shaped in cross section and includes first and second parts that are perpendicular one another, said first part being spaced from said interior surface and extending between said lateral wall and said second part, said second part being applied directly onto said interior surface; said first part defines a space with said interior surface that is filled in with said over-molded material; said over-molded material extends at least between said exterior and lateral walls and said interior surface; said interior wall faces said guiding mean and an interior surface of said movable window pane; said interior wall carries a second sealing lip which is configured to be in sealing contact with said guiding mean; said interior wall carries a third sealing lip which is configured to be in sealing contact with said movable window pane; said exterior wall is coated with an antifriction layer inside said longitudinal groove; said lateral wall is coated with an antifriction layer inside said longitudinal groove; said interior wall is coated with an antifriction layer inside said longitudinal groove; said module further comprises a second longitudinal profile which is U-shaped in cross section and which has an exterior wall connecting therebetween two lateral walls, said second profile defining a longitudinal groove configured to receive a metallic flange of the motor vehicle; said exterior wall of said second profile is parallel to said interior surface and is secured thereto by an over-molded material, one of said lateral walls of said second

profile being substantially aligned with a portion of said peripheral edge in a plane that is perpendicular to said fixed window pane, and the other of said lateral walls of said second profile extending perpendicularly and facing said interior surface; said exterior wall of said second profile is spaced from said interior surface; said module further comprises a third longitudinal profile which connects therebetween first and second longitudinal profiles; said module further comprises a third longitudinal profile which extends along at least an upper portion of said peripheral edge of said fixed window pane, and which is adjacent to said profile; said third longitudinal profile comprises first and second U-shaped portions, said first U-shaped portion being configured to receive a metallic flange of the motor vehicle and the second U-shaped portion being configured to receive by sliding said movable window pane; said first and second U-shaped portions define respectively two longitudinal grooves which are perpendicular one another; said over-molded material further comprises a third portion which is connected to the first portion and which is interposed between said leg and said peripheral edge; said over-molded material further comprises a third portion which is connected to the first portion, said third portion extending perpendicularly to the first portion and on said peripheral edge; said third portion includes an end which is coplanar with said exterior surface of said fixed window pane; the (co-extruded) material of the tooth is different from the (co-extruded) material of the first longitudinal profile; the (co-extruded) material of the protruding lip is different from the (co-extruded) material of the first longitudinal profile; the (co-extruded) material of the protruding leg is different from the (co-extruded) material of the first longitudinal profile; the material(s) of the tooth, and/or the protruding leg, and/or the protruding lip is/are softer or less hard than the material of the first longitudinal profile and in particular a covering material of this first longitudinal profile.

(21) The invention further proposes a motor vehicle, comprising at least one encapsulated fixed window module as defined above.

(22) The motor vehicle may comprise a door including a door frame, a movable window pane, and said encapsulated fixed window module, said movable window pane having an exterior surface, an interior surface, and a peripheral edge extending between said exterior and interior surfaces, wherein said exterior surfaces of said encapsulated fixed window module and of said movable window pane are substantially coplanar.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The invention will be better understood and other details, features and advantages of the invention will appear more clearly on reading the following description given by way of non-limiting examples and with reference to the accompanying drawings in which:

(2) FIG. 1 is a cross sectional view of adjacent window panes including a fixed window pane carrying a division post, and a movable window pane guided into a groove of the division post;

(3) FIG. 2 is a cross sectional view of adjacent and flush window panes including a fixed window pane carrying a division post, and a movable window pane carrying a guiding mean engaged into a groove of the division post;

(4) FIG. 3 is a fragmentary side view of an encapsulated fixed window module;

(5) FIG. 4 is a cross sectional view along line A1-A1 of FIG. 3;

(6) FIG. 5 is a cross sectional view along line A2-A2 of FIG. 3;

(7) FIG. 6 is a cross sectional view along line C-C of FIG. 3;

(8) FIG. 7 is a cross sectional view along line B-B of FIG. 3;

(9) FIG. 8 is a cross sectional view similar to FIGS. 1 and 7 and showing a first embodiment of the encapsulated fixed window module according to the invention;

(10) FIG. 9a is an enlarged view of a detail of FIG. 8;

(11) FIG. **9b** is an enlarged view of another detail of FIG. **8**; and
(12) FIGS. **10a** to **10i** are views corresponding to the view of FIG. **8** and showing further embodiments of the invention.

MORE DETAILED DESCRIPTION

(13) FIGS. **1** and **2** have been already described and illustrate the prior art.

(14) FIG. **3** shows an example of an encapsulated fixed window module **10** for a motor vehicle.

(15) A motor vehicle door comprises a door frame provided with an encapsulated fixed window module **10** which can be adjacent to a movable window pane **12**. In the example of FIG. **1**, a rear door of a vehicle is shown in which case the pane **12** is a forward pane and the module **10** is a rearward module. Another configuration would be on a front door, with in that case the movable window pane **12** positioned rearwardly from the module **10**.

(16) The module **10** includes at least a fixed window pane **14** molded on up to four extremities or along four edges. The or each molding forms an encapsulation **16**.

(17) Each pane **12**, **14** has an exterior surface **12a**, **14a** intended to be located outside the motor vehicle, an interior surface **12b**, **14b** intended to be located inside the motor vehicle, and a peripheral edge **12c**, **14c** extending between said exterior and interior surfaces (see FIGS. **4** and **6**).

(18) Each pane **12**, **14** may be substantially flat. Thus, its surfaces **12a**, **14a**, **12b**, **14b** may be substantially flat. Of course, the pane **12**, **14** and its surfaces may be slightly curved according to the expected design of the module. As shown in FIGS. **4** to **6**, the peripheral edge **12c**, **14c** may have a cross sectional convex shape.

(19) In the example shown, each pane **12**, **14** has a polygonal shape and comprises four longitudinal edges. Each peripheral edge **12c**, **14c** includes therefore a top edge **12ca**, **14ca**, a front edge **12cb**, **14cb**, a bottom edge **12cc**, **14cc** and a rear edge **12cd**, **14cd** (see FIG. **3**).

(20) The door may further comprise an appliqué **13**, the movable window pane **12** being located between the stationary window pane **14** and the appliqué **13**. The exterior surface **12a** of the pane **12** is advantageously flush with the exterior surface **14a** of the pane **14** and the exterior surface **13a** of the appliqué **13**.

(21) FIG. **4** is a cross sectional view at the top edge **12ca** of the movable window pane. FIG. **5** is a cross sectional view at the top edge **14ca** of the fixed window pane. FIG. **6** is a cross sectional view at the rear edge **14cd** of the fixed window pane. FIG. **7** is a cross sectional view at the front edge **14cb** of the fixed window pane and at the rear edge **12cd** of the movable window pane **12**.

(22) These figures show that the encapsulation **16** extends along the edges **14ca**, **14cb** and **14cd** of the pane **14**. Glass edge **14ca** and the inboard side of the glass next to **14cb** are encapsulated. Encapsulation of glass edges **14cd** and **14cc** is optional.

(23) Thus, the encapsulation **16** extends along at least two edges of the pane **14** in the example shown but may not necessarily cover these edges as will be detailed below. The encapsulation **16** may comprise segments or profiles or division posts each extending along a portion of the longitudinal edges of the pane **14**. Each profile of the division post has a length and a width, its length being greater than its width and extending along said portion of the longitudinal edges of the pane **14**.

(24) FIG. **4** shows a longitudinal profile which is called "third" profile **18**. This third profile **18** extends along said edges **14ca** and **12ca**. The portion of the profile **18** shown in FIG. **4** is the one extending along the edge **12ca** and is not over-molded.

(25) A substantially vertical profile (not shown) is molded to profile **18** to be part of module **10**, in order to guide the moveable glass edge **12cb**.

(26) The third profile **18** shown in FIG. **4** has first and second U-shaped portions **18a**, **18b** in cross section.

(27) The first U-shaped portion **18a** defines a longitudinal groove **24** that is oriented inwardly and is configured to receive a metallic flange **26** of the motor vehicle. In the example shown, the metallic flange includes two substantially horizontal panels that are superimposed and inserted into

the groove **26**. The first portion **18a** includes an exterior wall **18aa** connecting two lateral walls **18ab**, **18ac** defining the groove **26** therebetween.

(28) The second U-shaped portion **18b** defines a longitudinal groove **28** that is oriented downwardly in a plane parallel to the panes **12**, **14** and which is configured to receive by sliding an upper portion of the movable window pane **12**. The second portion **18b** includes an interior wall **18ba** and an exterior wall **18bb** connected by a lateral wall which may be integrated in said lateral wall **18ac** as shown in the figure. The walls of the second portion **18b** define said groove **28**.

(29) Both portions **18a**, **18b** include sealing lips intended to cooperate with the pane **12** and/or parts of the vehicle.

(30) The third profile **18** further may include an elongated esthetic component **30** which extends along the profile **18** and is secured onto both portions **18a**, **18b**. The component **30** includes an esthetic exterior face **30a** and an interior face **30b**. The component **30** includes a substantially flat portion including the faces **30a**, **30b** and longitudinal hooks **30c** extending inwardly from the inner face **30b**. The hooks **30c** are used to mechanically secure the component **30** onto the profile **18**. The component **30** is intended to cover at least portions of said exterior walls **18aa**, **18bb** which both comprise grooves intended to receive the hooks **30c** by snap fitting.

(31) FIG. 5 still shows the third profile **18** and in particular its portion extending along the edge **14ca** and is over-molded.

(32) This portion of the profile **18** differs from the portion of FIG. 2 in that it does not comprise the interior wall **18ba** and in that it is secured to the fixed window pane **12** by an over-molded material **50**.

(33) This over-molded material **50** extends onto the inner surface of the exterior wall **18bb**, the downward surface of the lateral wall **18ac** and also over the edge **14ca** and portions of the surfaces **14a**, **14b** of the fixed window pane **14**. This over-molded material **50** may have lips or legs intended to abut against parts of the vehicle.

(34) FIG. 6 shows a longitudinal profile which is called “second” profile **20**. This second profile **20** extends along said edge **14cd** and is over-molded.

(35) The second profile **20** shown in FIG. 6 is U-shaped in cross section and defines a longitudinal groove **32** that is oriented inwardly and is configured to receive a metallic flange **34** of the motor vehicle. In the example shown, the metallic flange **34** includes two substantially vertical panels that are superimposed and inserted into the groove **32**.

(36) The second profile **20** includes an exterior wall **20a** and two lateral walls **20b**, **20c** connected by said exterior wall **20a**. The exterior wall **20a** is usually parallel to the interior surface **14b** of the fixed window pane **14** and is secured thereto by an over-molded material **36**. In the example shown, the exterior wall **20a** faces the interior surface **14b** and spaced therefrom. This space **S** is filled in with said over-molded material **36**.

(37) The lateral wall **20b** is substantially aligned with a portion (the rear edge **14cd** in the example shown) of said peripheral edge **14c** in a plane **P1** that is perpendicular to the fixed window pane **14**. The lateral wall **20c** extends perpendicularly to and faces the interior surface **14b**.

(38) In the example shown, the over-molded material **36** further extends over the lateral wall **20c** and the front edge **14cd**. The lateral wall **20b** is free of such over-molded material **36**. Said over-molded material **36** is V-shaped (with an angle lower than 90°) in front of the rear edge **14cd** and defines a first planar surface **36a** aligned with the exterior surface **14a** and a second planar surface **36b** which extends between the first surface **36a** and the profile **20** and is inclined with respect to said plane **P1**.

(39) The second profile **20** includes sealing or retention lips intended to cooperate with parts of the vehicle.

(40) FIG. 7 shows another longitudinal profile which is called “first” profile **22**, also known as division post. This first profile **22** extends substantially vertically along said edges **12cd**, **14cb** and is over-molded.

(41) The first profile **22** shown in FIG. 7 is U-shaped in cross section and defines a longitudinal groove **38** that is oriented forwardly and is configured to receive a guiding mean or guide **40** secured to the movable window pane **12**.

(42) The first profile **22** includes an exterior wall **22a** and an interior wall **22b** connected together by a lateral wall **22c** and defining the groove therebetween. The exterior wall **22a** is parallel to the interior surface **14b** of the fixed window pane **14** and is secured thereto by an over-molded material **42**. In the example shown, the exterior wall **22a** faces the interior surface **14b** and is L-shaped in cross section. The exterior wall **22a** includes first and second parts **22aa**, **22ab** that are perpendicular to one another, said first part **22aa** being spaced from the interior surface **14b** and extending between said lateral wall **22c** and said second part **22ab**. Said second part **22ab** is applied directly onto said interior surface **14b**. Said first part **22ab** defines a space S with said interior surface **14b** that is filled in with the over-molded material **42**.

(43) The over-molded material **42** further extends over at least a portion of said lateral wall **22c**. The over-molded material **42** does not cover the peripheral edge **14cb** along said first profile **22**, i.e., the front edge **14cb**, in the example shown.

(44) The exterior wall **22a** has a longitudinal edge **22a1** which is opposite to the lateral wall **22c** that is substantially aligned with said front edge **14cb** in a plane P2 that is perpendicular to the fixed window pane **14**.

(45) This longitudinal edge **22a1** carries a first sealing lip **44** which is configured to abut against both edges **14cb** and **12cd**. The sealing lip **44** is the sole member extending between said edges **14cb** and **12cd**. The sealing lip **44** is curved (concave or convex) and defines a groove oriented inwardly. The lip **44** includes a longitudinal connecting edge **44a** which extends in the plane P2 and is secured to the longitudinal edge **22a1**, and further includes an opposite longitudinal sealing edge **44b** abutting against the rear edge **12cd** of the pane **12**. The longitudinal portion of the sealing lip **44** extending between both edges **44a**, **44b** has an exterior surface **44c** which is substantially aligned with said exterior surfaces **14a**, **12a**.

(46) As far as the interior wall **22b** of the first profile **22** is concerned, it crosses said plane P2 and is configured to face said guiding mean **40**, and preferably also to face the interior surface **12b** of the movable window pane **12** as shown in the figure.

(47) The interior wall **22b** carries a second sealing lip **46** which is configured to be in sealing contact with the guiding mean **40**. In another example, only the first and the third sealing lips may be provided (no sealing is then provided on the guiding mean **40**).

(48) The guiding mean **40** may be formed of a single longitudinal part that is secured, for instance by gluing or over-molding, onto the interior surface **12b** of the pane **12** and includes a guiding leg **40a** crossing the plane P2 and received into the groove **38** for sliding motion. As shown in the drawings, the surfaces of the groove **38** may be coated with an antifriction layer suitable to cooperate by sliding with the guiding mean **40** and in particular its leg **40a**. Therefore, as shown in the drawings, the walls **22a**, **22b**, **22c** may be coated with the antifriction layer inside the groove **38**.

(49) As shown in FIGS. 4 to 7, each profile **18**, **20**, **22** may comprise at least one U-shaped metallic rail embedded or encapsulated into a polymeric covering. The rail includes an interior branch and an exterior branch, connected together by a lateral branch. Each profile and its polymeric covering may be made by co-extrusion. The sealing lips may be made of an elastically deformable material, such as for example EPDM or thermoplastic. These lips may be obtained by over-molding during a process where the profiles **18**, **20**, **22** are connected together in their interconnection zones. Of course, each profile might not comprise metallic rail and might therefore comprise only polymeric materials. Each profile may have polypropylene in areas where material rigidity is needed and the lips might be made of a softer material. The profiles are secured onto the fixed window pane **14** by over-molding by means of the over-molded material **36**, **42** and **50**. Stationary glass edge **14cc** is usually encapsulated as well.

(50) FIG. 8 is a view similar to the view of FIGS. 2 and 7 and shows a first embodiment according to the invention.

(51) The invention relates to an encapsulated fixed window module which is similar to the encapsulated window module described above in relation with FIGS. 3 to 6. The above description made in reference to FIGS. 3 to 6 applies therefore to the claimed invention and the description of FIG. 7 of the former embodiment may be replaced by the following description of FIG. 8 which illustrates the first embodiment of the invention.

(52) The reference numerals used in relation to FIGS. 3 to 7 are used for designating the same elements in FIG. 8.

(53) The window panes 12 and 14 are similar to the ones disclosed above. The guiding mean 40 is also similar.

(54) The (first) profile 122 or division post of FIG. 8 extends along the edges 12cd, 14cb and is over-molded.

(55) The profile 122 shown in FIG. 7 is U-shaped in cross section and defines a longitudinal groove 38 that is oriented forwardly and is configured to receive the guiding mean 40 secured to the movable window pane 12.

(56) The profile 122 includes an exterior wall 22a and an interior wall 22b connected together by a lateral wall 22c and defining the groove 38 therebetween. The exterior wall 22a is parallel to the interior surface 14b of the fixed window pane 14 and is secured thereto by an over-molded material 142.

(57) As shown in the drawings, the surfaces of the groove 38 may be coated with an antifriction layer F1 suitable to cooperate by sliding with the guiding mean 40 and in particular its leg 40a. Therefore, as shown in the drawings, the walls 22a, 22b, 22c may be coated with the antifriction layer F1 inside the groove 38.

(58) The exterior wall 22a has a surface 22a1 that faces the interior surface 14b and is L-shaped in cross section. The exterior wall 22a includes first and second parts 22aa, 22ab that are perpendicular one another, said first part 22aa being spaced from the interior surface 14b and extending between said lateral wall 22c and said second part 22ab. Said second part 22ab is applied directly onto said interior surface 14b in the vicinity of the edge 14c, 14cb.

(59) Said second part 22ab carries a protruding leg 160 which extends externally in a direction substantially perpendicular to the exterior wall 22a. The protruding leg 160 is interposed in the gap between both peripheral edges 12cb, 14cb, and is in abutment against the peripheral edge 14cb. The protruding leg 160 carries a first sealing lip 44 which is configured to abut against the edge 12cb. The first sealing lip 44 has an exterior surface 44c which is substantially aligned with the exterior surfaces 14a, 12a.

(60) The protruding leg 160 is made of the same material as the main material (or polymeric covering) of the profile 122. The first sealing lip 44 is preferably made of a softer material.

(61) The exterior wall 22a, and in particular its first part 22aa, defines a space S with the interior surface 14b that is filled in with the over-molded material 142 and in particular with a first portion 142a of this over-molded material 142.

(62) The over-molded material 142 further extends over at least a portion of said lateral wall 22c and comprises a second portion 142b extending between the lateral wall 22c and the interior surface 14b. The over-molded material 142 does not cover the peripheral edge 14cb in the example shown.

(63) As far as the interior wall 22b of the first profile 22 is concerned, it is longer than the exterior wall 22a and is configured to face said guiding mean 40, and preferably also to face the interior surface 12b of the movable window pane 12 as shown in the figure.

(64) The interior wall 22b carries a second sealing lip 46 which is configured to be in sealing contact with the guiding mean 40, and a third sealing lip 48 which is configured to be in sealing contact with the interior surface 12b of the movable window pane 12.

(65) Sealing lips **44**, **46** and **48** may be coated with antifriction layers F1.

(66) In another example, only the first and the third sealing lips may be provided (no sealing is then provided on the guiding mean **40**).

(67) The profile **122** may comprise at least one U-shaped metallic rail embedded or encapsulated into a polymeric covering. This rail may include a first branch extending into the exterior wall **22b**, a second branch extending into the interior wall **22a**, and a connecting branch extending into the lateral wall **22c**.

(68) The profile **122** and its polymeric covering may be made by co-extrusion. The sealing lips may be made of an elastically deformable material, such as for example EPDM or thermoplastic. These lips may be realized by over-molding or co-extrusion. Of course, the profile **122** might not comprise metallic rail and might therefore comprise only a polymeric material. The polymeric or covering material of the profile **122** may be a thermoplastic or an elastomer thermoplastic, such as for instance polypropylene. The over-molded material is for instance made of thermoplastic, elastomer thermoplastic, EPDM, SEBS, etc.

(69) As shown in FIG. **8**, the first portion **142a** of the over-molded material **142** has a first thickness E1 which is constant along a major portion of this first portion. The thickness E1 of this first portion **142a** is measured in a direction which is perpendicular to the interior surface **14b** of the fixed window pane **14** and which is also perpendicular to the exterior wall **22a**.

(70) The second portion **142b** of the over-molded material **142** has a second thickness E2 which is constant along a major portion of the second portion. The thickness E2 of this second portion **142b** is measured in a direction which is perpendicular to the lateral wall **22c**.

(71) This second portion **142b** has an outer face **142b1** which extends between the lateral wall **22c** and the interior surface **14b** and which includes a major portion which is inclined with respect to the interior surface **14b**.

(72) In the shown example, the over-molded material **142** is V-shaped and includes both portions **142a**, **142b** connected therebetween and forming together a tip **142c**. The over-molded material **142** includes a protruding rib **166** on its tip **142c**, this protruding rib **166** extending on the interior surface **14b** (FIG. **9a**).

(73) In the example shown, the shape of the over-molded material is provided by the shape of the profile **122** which includes a tooth **168** or a fin (such as a shark fin) on its lateral wall **22c**.

(74) The tooth **168** includes a first surface **168a** parallel to the interior surface **14b** and aligned with the surface **22a1** of the exterior wall **22a**, and a second surface **168b** which is inclined with respect to the interior surface **14b**. The function of tooth **168** is to create a constant offset between the second surface **168b** and the outer face **142b1**. The shape and location of the surface **168b** will thus evolve according to the shape and location of the outer face **142b1**. The thickness E2 of the second portion **142b** may be measured in a direction which is perpendicular to this second surface **168b**.

(75) The tooth **168** is preferably fully embedded into the over-molded material so that the first portion **142a** extends between the first surface **168a** and the interior surface **14b** and the second portion **142b** extends between the second surface **168b** and the interior surface **14b**. The first surface **168a** is planar and the second surface **168b** is also planar in the example shown. This second surface **168b** is preferably parallel to the outer face **142b1** to ensure that E2 is constant. The first and second surfaces **168a**, **168b** are therefore V-shaped in cross section in the example shown. The first and second surfaces **168a**, **168b** form an angle α or form a tip having an angle which is comprised between 20 and 85°, and preferably between 50 and 70°. In other words, the tooth **168** defines an angle α comprised between 20 and 85°, and preferably between 50 and 70°.

(76) FIGS. **10a** to **10e** show further embodiments of the invention.

(77) The embodiment of FIG. **10a** differs from the embodiment of FIG. **8** in that the second part **22ab** of the exterior wall **22a** is replaced by a protruding lip **162** which extends externally and which is in abutment against the interior surface **14b** of the window pane **14**.

(78) The protruding lip **162** and the protruding leg **160** may be made of the same material as the

main material of the profile **122**. The first sealing lip **44** is made of a softer material.

(79) The over-molded material **142** does not cover the peripheral edge **14cb**. This is due to the presence of the protruding lip **162** which abuts against the pane **14** and avoid over-molded material **142** passing through the lip **162** and up to the edge **14cb**.

(80) As shown in the drawings, the protruding lip **162** and leg **160** may define together and with the pane **14** a space S' that is empty, i.e., that is not filled with over-molded material **142**.

(81) The embodiment of FIG. **10b** differs from the embodiment of FIG. **8** in that the tooth **168** defines a larger angle α of about 70-85°. The V-shape of the over-molded material **142** has a similar angle.

(82) The embodiment of FIG. **10c** differs from the embodiment of FIG. **8** in that the tooth **168** defines a small angle α of about 20-40°. The V-shape of the over-molded material **142** has a similar angle.

(83) The embodiment of FIG. **10d** differs from the embodiment of FIG. **8** in that the second surface **168b** of the tooth **168** is curved and in particular is concave. The second portion **142b** of the over-molded material **142** conforms to the shape of this second surface **168b** and has a curved shape. Similarly, the outer face **142b1** of the over-molded material **142** is curved and concave to get a thickness $E2$ as constant as possible.

(84) The embodiment of FIG. **10e** differs from the embodiment of FIG. **8** in that the second surface **168b** of the tooth **168** is curved and in particular convex. The second portion **142b** of the over-molded material **142** conforms to the shape of this second surface **168b** and has a curved shape. Similarly, the outer face **142b1** of the over-molded material **142** is curved and convex to get a thickness $E2$ as constant as possible.

(85) The embodiment of FIG. **10f** differs from the embodiment of FIG. **8** in that the metallic rail embedded or encapsulated into the polymeric covering includes a leg **170** at its edge opposite the connecting branch. This leg **170** extends outwardly and is interposed between the edges **12cd** and **14cb**.

(86) The embodiment of FIG. **10g** differs from the embodiment of FIG. **8** in that over-molded material **142** comprises a third portion **142c**, the first portion **142a** extending between the third and second portions **142c**, **142b**.

(87) The third portion **142c** extends on the peripheral edge **14cb** of the fixed pane **14**. The third portion **142c** is perpendicular to the first portion **142a** and extends outwardly from the end of the first portion **142a** which is opposite to the second portion **142b**. The third portion **142c** covers entirely the peripheral edge **14cb** in the example shown.

(88) Therefore, the profile **122** is not directly in contact with the pane **14**. Its exterior wall **22b** is remote from the interior surface **14b** and is separated therefrom by the over-molded material **142**.

(89) The first sealing lip **44** of the profile **122** is configured to abut against edge **12cd**. The third portion **142c** and the sealing lip **44** are both located between the edges **14cb** and **12cd**.

(90) As shown, the third portion **142c** may include an end which is coplanar with said exterior surfaces of said panes **12**, **14**.

(91) The embodiment of FIG. **10g** further differs in that the antifriction layer on the exterior surface of the interior wall **22a** of the profile **122** is replaced by a co-extruded thermoplastic material and/or a cellular or foam material $F2$. This material is used to prevent rattle noise/tapping noise during door slam. The material $F2$ has for instance a shore A hardness of about 25-60 and/or a coefficient of friction less than about 0.6. All the embodiments described above may have such material in the groove **38** instead of one of the antifriction layers $F1$.

(92) In another embodiment shown in FIG. **10h**, the third portion **142c** of FIG. **10g** may be combined with the leg **160** of any of the preceding embodiments (FIG. **8** to FIG. **10g**) and may therefore be interposed between the peripheral edge **14cb** of the pane **14** and this leg **160**.

(93) Generally speaking, the third portion **142c** is advantageous because it allows simplifying the mold that is used to perform the encapsulation. Moreover, it enhances the sealing during molding at

the connecting area between the edges **14cb**, **14ca** (cf. FIG. 3).

(94) In another embodiment shown in FIG. **10i**, the tooth **168** is made of a material which is different from the main (co-extruded or covering) material of the profile **122**. For instance, the main material of the profile **122** is harder than the material of the tooth **168**. The main material is for instance a PP. The material of the tooth **168** may have a Shore A hardness of between 70-95, and for instance of about 85. The material of the tooth **168** is for instance a TPE.

(95) The protruding leg **160** and the protruding lip **162** may also be made of a material which is different from the main material of the profile **122**. The materials of the leg **160**, of the lip **162** and of the tooth **168** may be the same.

(96) Even if the tooth **168** and/or the protruding leg **160** and/or the protruding lip **162** may be made of different material, they are preferably made by co-extrusion with the profile **122**. Use of a softer material for the protruding leg **160** for instance enhances the molding operation.

(97) The second portion **142b** extends on the tooth **168** and has an end linked to the profile **122**. This portion **142b** has a constant thickness E2 up to this end.

(98) In the embodiment of FIG. **10i**, both branches of the metallic rail of the profile **122** are straight while in the embodiments of FIGS. **10a-10h** the exterior branch of the metallic rail is slightly inclined and has an end opposite its lateral branch which protrudes inwardly.

(99) In all the embodiments, the esthetic appearance of the window module may be enhanced by lying flush the outside surfaces **12a**, **14a** of the fixed and movable window panes **12**, **14**. The exterior surfaces **12a**, **14a** of the panes **12**, **14** may be therefore substantially coplanar. These surfaces **12a**, **14a** may also be flush with the exterior surface **44c** (cf. FIGS. **8** to **10f**), and/or with the end of the leg **160** (FIGS. **8** to **10f**), and/or with the end of the third portion **142c** of the over-molding material **142** (FIGS. **10g** and **10h**).

Claims

1. An encapsulated fixed window module for a motor vehicle, comprising: a fixed window pane having an exterior surface, an interior surface, and a peripheral edge extending between said exterior and interior surfaces, at least one division post extending along at least a portion of said peripheral edge, said division post comprising a first longitudinal profile which is U-shaped in cross section and which has an exterior wall, an interior wall, and a lateral wall connecting said exterior and interior walls, said first profile defining a longitudinal groove configured to receive a guiding mean carried by a movable window pane, said exterior wall facing said interior surface, and an over-molded material securing said first profile to said fixed window pane, said over-molded material comprising a first portion extending between said exterior wall and said interior surface and a second portion extending between said lateral wall and said interior surface, said first and second portions being connected one to the other, said first portion having a first thickness E1 which is constant along a major portion of said first portion, said second portion having an outer face which extends between the lateral wall and said interior surface and which includes a major portion which is inclined with respect to said interior surface, wherein said second portion has a second thickness E2 which is constant along a major portion of said second portion.

2. The encapsulated fixed window module as claimed in claim 1, wherein said lateral wall comprises a tooth which includes a first surface parallel to said interior surface, and a second surface which is inclined with respect to said interior surface, said first portion extending between said first surface and said interior surface and said second portion extending between said second surface and said interior surface.

3. The encapsulated fixed window module as claimed in claim 2, wherein said first surface is planar and said second surface is planar, concave or convex.

4. The encapsulated fixed window module as claimed in claim 2, wherein said second surface is mainly parallel to said outer surface or provides a constant offset when this second surface is

concave or convex.

5. The encapsulated fixed window module as claimed in claim 2, wherein said first and second surfaces are V-shaped in cross section.

6. The encapsulated fixed window module as claimed in claim 2, wherein said tooth has a tip having an angle comprises between 2° and 85° .

7. The encapsulated fixed window module as claimed in claim 1, wherein said over-molded material is V-shaped.

8. The encapsulated fixed window module (10) as claimed in claim 1, wherein

$0.8 \leq E1 \leq E2 \leq 1.2$, in which E1 and E2 are respectively said first and second thicknesses.

9. The encapsulated fixed window module as claimed in claim 1, wherein said exterior wall comprises a longitudinal end opposite said lateral wall, said longitudinal end comprising a protruding leg which is perpendicular to the exterior wall and is in abutment against said peripheral edge.

10. The encapsulated fixed window module as claimed in claim 9, wherein said protruding leg carries a first sealing lip which is configured to abut against a peripheral edge of said movable window pane.

11. The encapsulated fixed window module as claimed in claim 9, wherein said longitudinal end and/or said exterior wall carries a protruding lip which abuts against said interior face and which is in contact with said over-molded material.

12. The encapsulated fixed window module as claimed in claim 11, wherein an empty space is defined between said fixed window pane on one hand, and said protruding leg and lip on the other hand.

13. The encapsulated fixed window module as claimed in claim 1, wherein said first profile comprises an embedded metallic rail which includes a leg extending outwardly and facing said peripheral edge.

14. The encapsulated fixed window module as claimed in claim 13, wherein said over-molded material further comprises a third portion which is connected to the first portion and which is interposed between said leg and said peripheral edge.

15. The encapsulated fixed window module as claimed in claim 1, wherein said over-molded material further comprises a third portion which is connected to the first portion, said third portion extending perpendicularly to the first portion and on said peripheral edge.

16. The encapsulated fixed window module as claimed in claim 15, wherein said third portion includes an end which is coplanar with said exterior surface of said fixed window pane.

17. A motor vehicle, comprising at least one encapsulated fixed window module as claimed in claim 1.

18. The motor vehicle as claimed in claim 17, which comprises a door including a door frame, a movable window pane, and said encapsulated fixed window module, said movable window pane having an exterior surface, an interior surface, and a peripheral edge extending between said exterior and interior surfaces, wherein said exterior surfaces of said encapsulated fixed window module and of said movable window pane are substantially coplanar.
